Data and Data Preprocessing

Lecture 2

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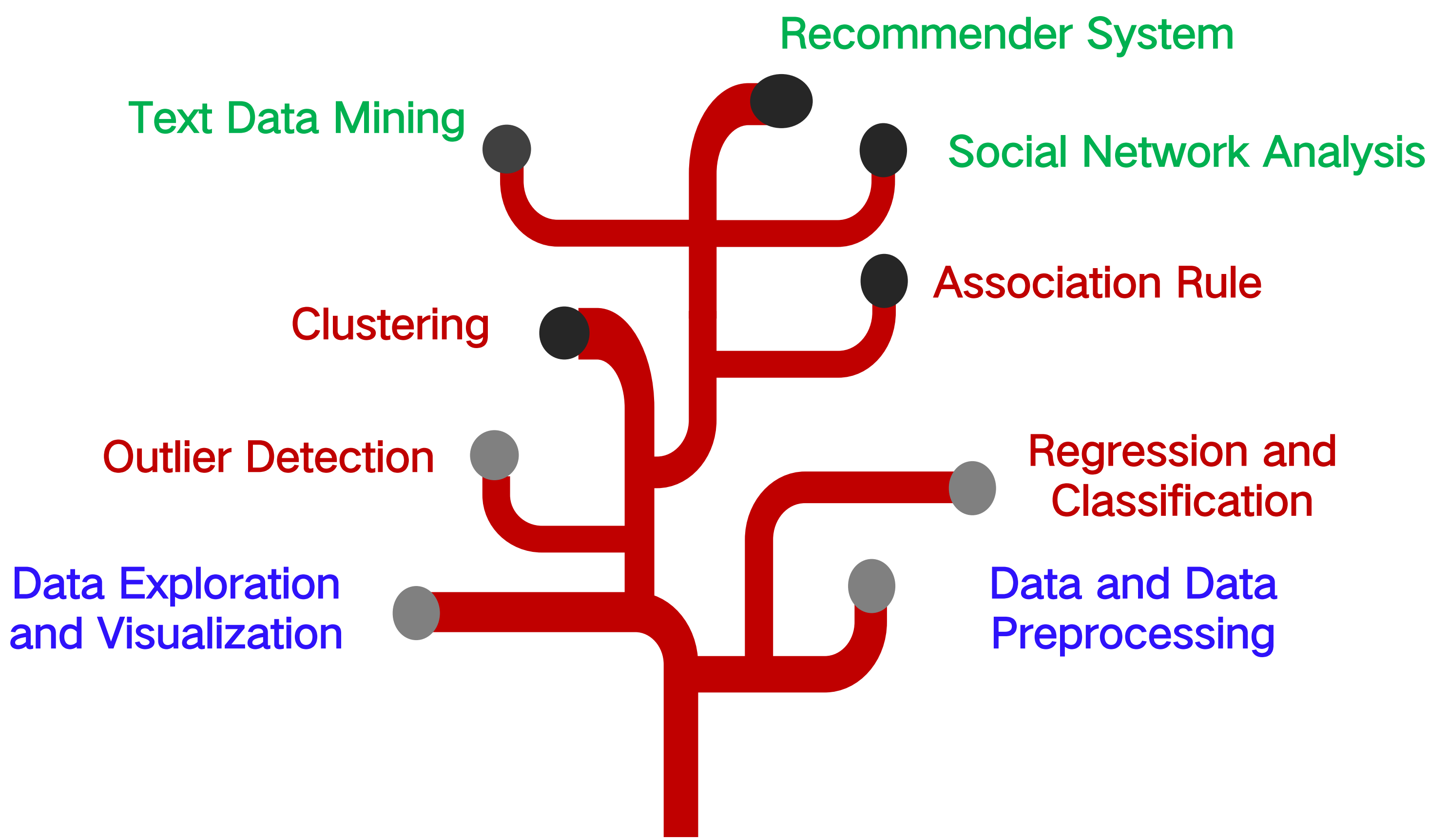
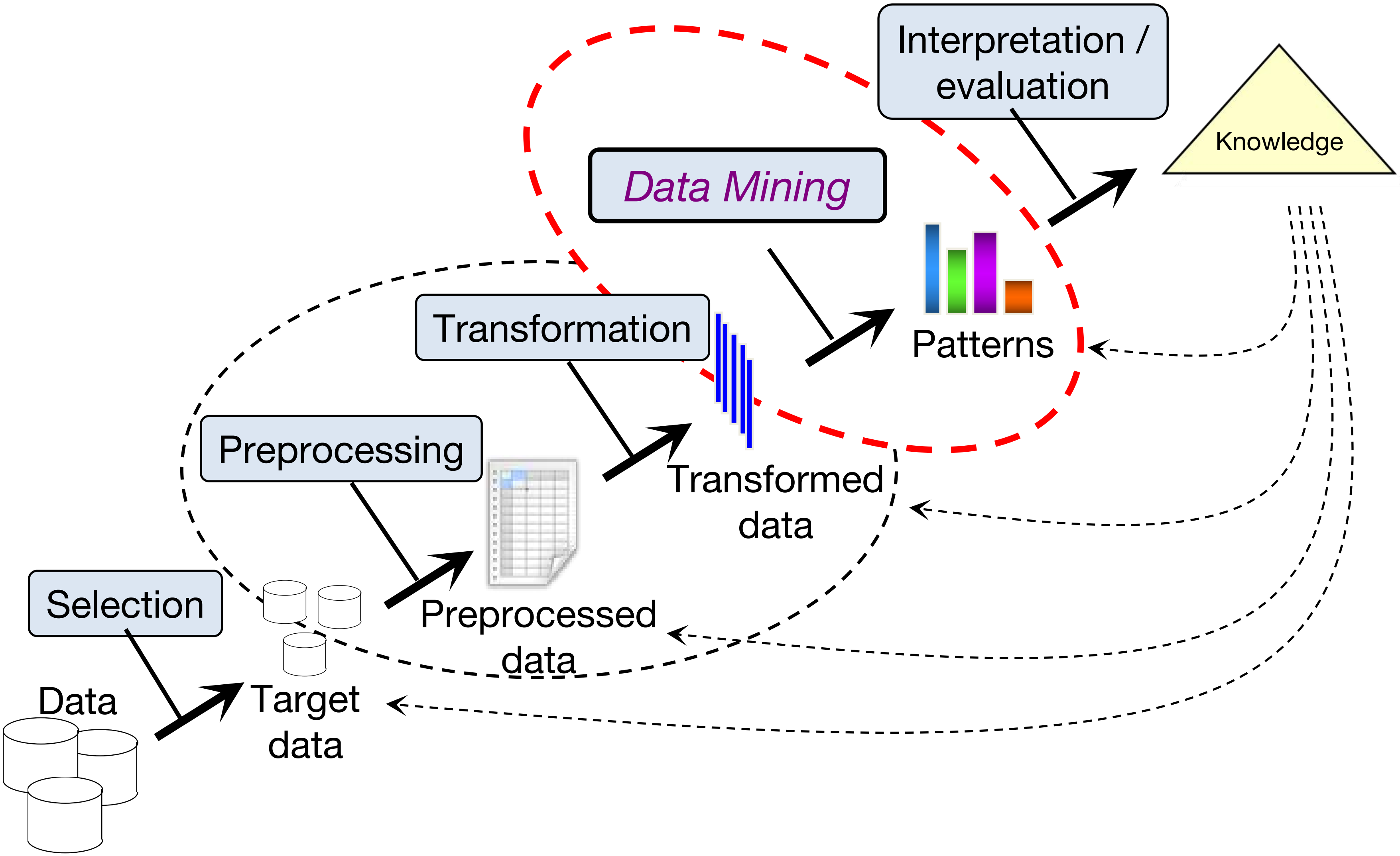
Our course tree



Our Data Mining Course Tree

Stages of Data Mining



Outline

- What is Data
- Data Quality Issues
- Data Preprocessing
  - Attribute transformation
  - Feature creation
  - Feature selection
  - Dimensionality reduction

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What is data?

- Collection of data objects and their attributes
- An attribute is a property or characteristic of an object
  - Examples: eye color of a person, temperature, etc.
  - Attribute is also known as variable, field, characteristic, or feature
- A collection of attributes describe an object
  - Object is also known as record, point, case, sample, entity, or instance

Attributes

Objects

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Attribute values

- Attribute values are numbers or symbols assigned to an attribute

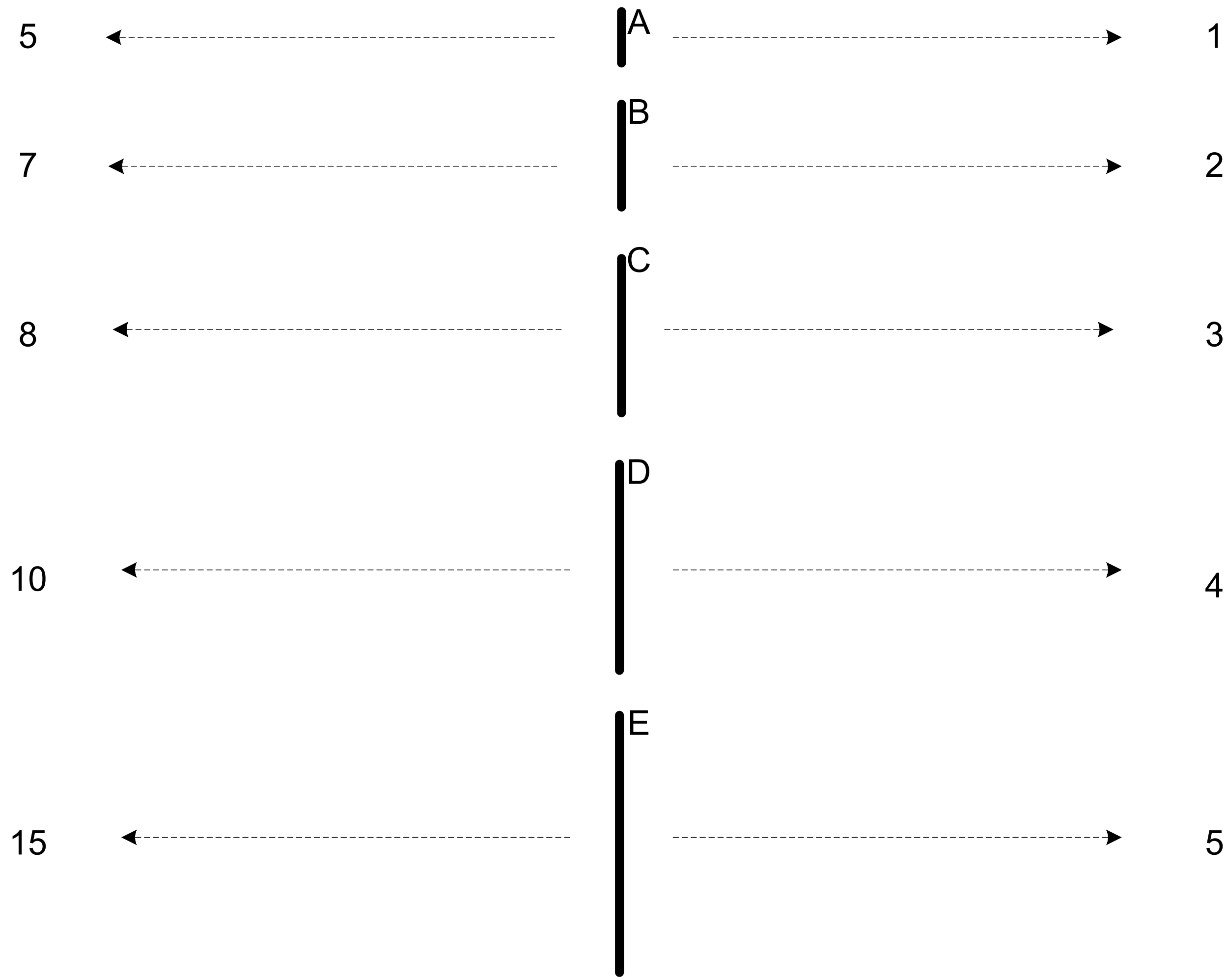
- Distinction between attributes and attribute values
  - Same attribute can be mapped to different attribute values
    - Example: height can be measured in feet or meters

  - Different attributes can be mapped to the same set of values
    - Example: Attribute values for ID and age are integers
    - But properties of attribute values can be different
      - ID has no limit but age has a maximum and minimum value

  - The way you measure an attribute may not match the attributes properties (here is an example)

Attribute values

This scale preserves only the ordering property of length.



This scale preserves the ordering and additvity properties of length.

Types of attributes

- There are different types of attributes
  - Nominal
    - Examples: ID numbers, eye color, zip codes
  - Ordinal
    - Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height in {tall, medium, short}
  - Interval
    - Examples: calendar dates, temperatures in Celsius or Fahrenheit.
  - Ratio
    - Examples: temperature in Kelvin, length, time, counts

Properties of attributes

- **The type of an attribute depends on which of the following properties it possesses:**
  – Distinctness: $= \ \neq$
  – Order: $< \ >$
  – Addition: $+ \ -$
  – Multiplication: $* \ /$

  – Nominal attribute: distinctness
  – Ordinal attribute: distinctness & order
  – Interval attribute: distinctness, order & addition
  – Ratio attribute: all four properties

Discrete and continuous attributes

- Discrete attribute
  - Has only a finite or countably infinite set of values
  - Examples: zip codes, counts, or the set of words in a collection of documents
  - Often represented as integer variables.
  - Note: binary attributes are a special case of discrete attributes

- Continuous attribute
  - Has real numbers as attribute values
  - Examples: temperature, height, or weight.
  - Practically, real values can only be measured and represented using a finite number of digits.
  - Continuous attributes are typically represented as floating-point variables.

Asymmetric Attributes

- Only presence (a non-zero attribute value) is regarded as important
  - Words present in documents
  - Items present in customer transactions

- If we met a friend in the grocery store would we ever say the following?

"I see our purchases are very similar since we didn't buy most of the same things."

Important Characteristics of Data

- Dimensionality (number of attributes)
  - High dimensional data brings a number of challenges
- Distribution Sparsity
  - Skewness
  - Sparsity (Only presence counts)
- Resolution
  - Patterns depend on the scale
- Size
  - Type of analysis may depend on size of data

Types of data sets

- Record
  - Data matrix
  - Document data
  - Transaction data

- Graph
  - World Wide Web
  - Molecular structures

- Ordered/Sequence
  - Spatial data
  - Temporal (time series) data
  - Sequential data
  - Genetic sequence data

Record data

- Data that consists of a collection of records, each of which consists of a fixed set of attributes

Tid	**Refund**	**Marital Status**	**Taxable Income**	**Cheat**
1	Yes	Single	125K	**No**
2	No	Married	100K	**No**
3	No	Single	70K	**No**
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6	No	Married	60K	**No**
7	Yes	Divorced	220K	**No**
8	No	Single	85K	**Yes**
9	No	Married	75K	**No**
10	No	Single	90K	**Yes**

Data matrix

- If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute.
- Such data set can be represented by an *m* x *n* matrix, where there are *m* rows, one for each object, and *n* columns, one for each attribute

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

Document data

- Each document becomes a ‘term’ vector,
  - each term is a component (attribute) of the vector
  - the value of each component is the number of times the corresponding term occurs in the document.

	team	coach	play	ball	score	game	win	lost	timeout	season
document 1	3	0	5	0	2	6	0	2	0	2
document 2	0	7	0	2	1	0	0	3	0	0
document 3	0	1	0	0	1	2	2	0	3	0

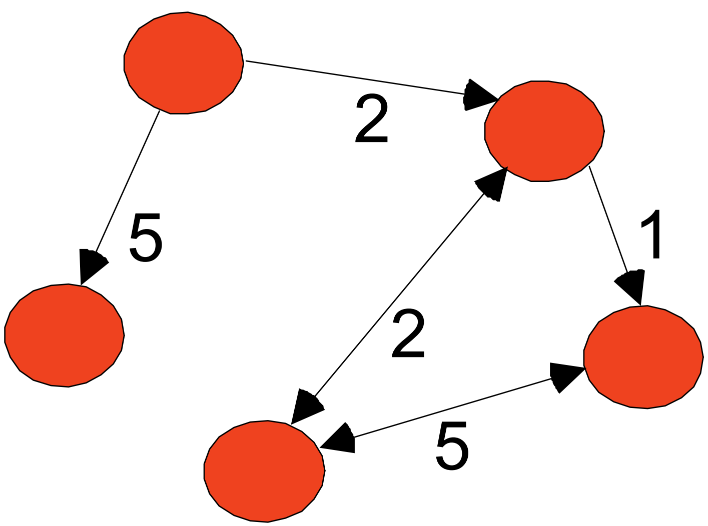
Transaction data

- A special type of record data, where
  - Each record (transaction) involves a set of items.
  - For example, consider a grocery store. The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.

TID	*Items*
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Graph data

- Examples: Generic graph and HTML Links



Data Mining

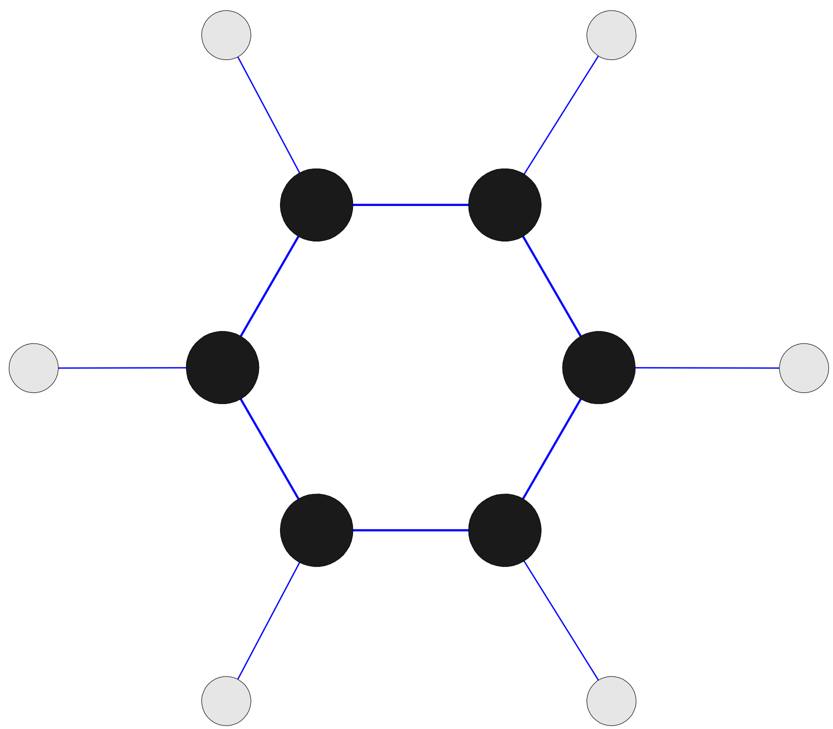
Graph Partitioning

Parallel Solution of Sparse Linear System of Equations

<a href="papers/papers.html#ffff">
N-Body Computation and Dense Linear System Solvers

Chemical data

- Benzene molecule: C_6H_6



Ordered data

- Sequences of transactions

Items/Events

(A B) (D) (C E)

(B D) (C) (E)

(C D) (B) (A E)

An element of
the sequence

Ordered data

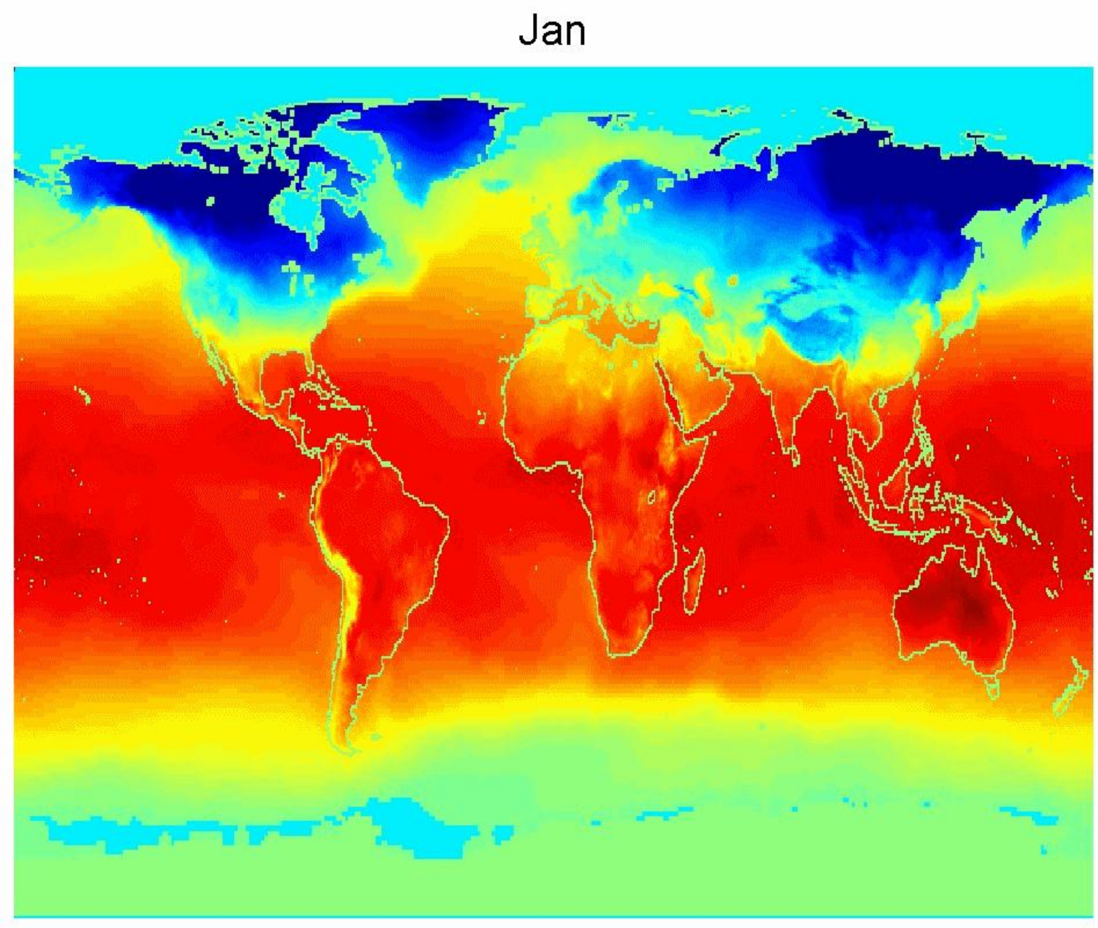
- Genomic sequence data

```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```

Ordered data

- Spatio-temporal data

Average monthly temperature of land and ocean



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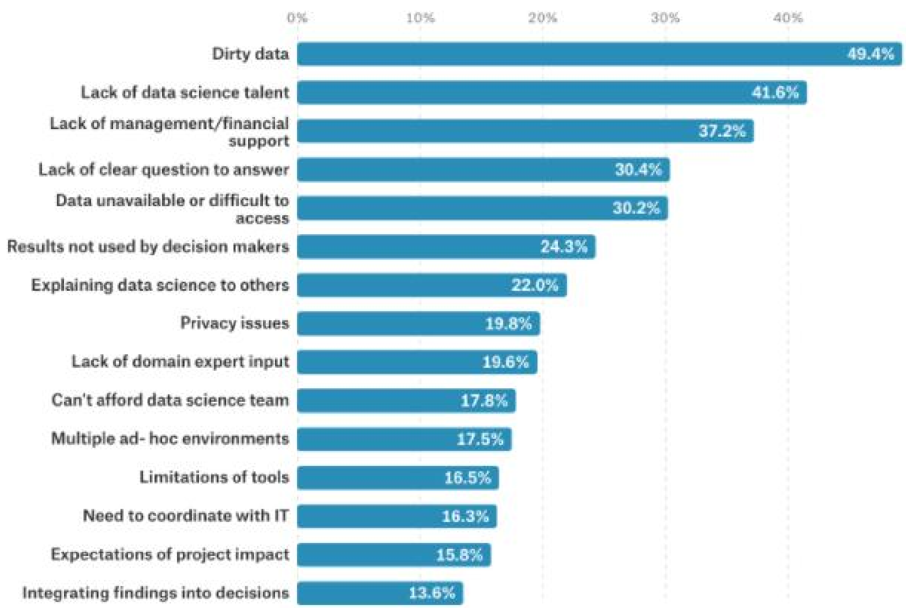
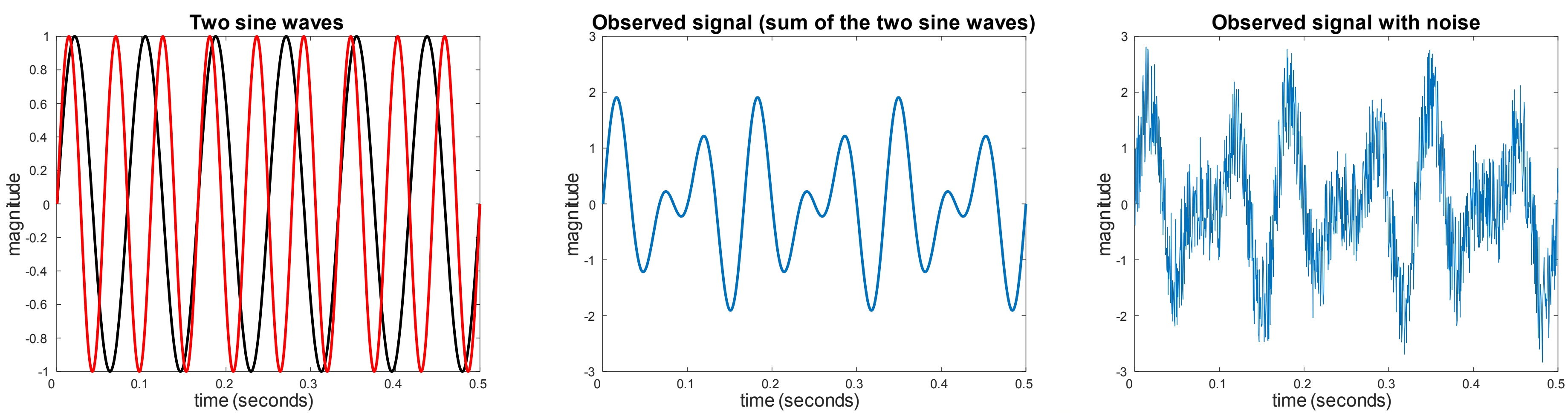
Data quality



- Examples of data quality problems:
  – noise and outliers
  – missing values
  – duplicate data

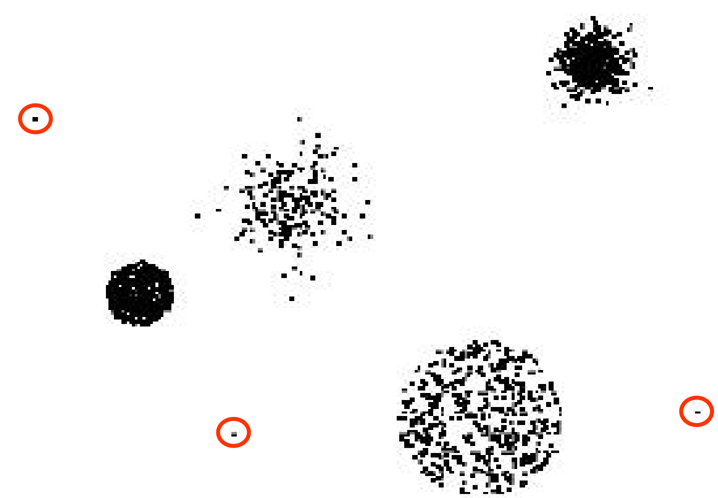
Noise

- For objects, noise is an extraneous object
- For attributes, noise refers to modification of original values
  - Examples: distortion of a person's voice when talking on a poor phone and "snow" on television screen
  - The figures below show two sine waves of the same magnitude and different frequencies, the waves combined, and the two sine waves with random noise (The magnitude and shape of the original signal is distorted)



Outliers

- ***Outliers*** are data objects with characteristics that are considerably different than most of the other data objects in the data set
  - **Case 1:** Outliers are noise that interferes with data analysis
  - **Case 2:** Outliers are the goal of our analysis
    - Credit card fraud
    - Intrusion detection



- Causes?

Missing values

- Reasons for missing values
  - Information is not collected
    (e.g., people decline to give their age and weight)
  - Attributes may not be applicable to all cases
    (e.g., annual income is not applicable to children)

- Handling missing values
  - Eliminate data objects
  - Estimate missing values (imputation)
  - Ignore the missing value during analysis
  - Replace with all possible values (weighted by their probabilities)

Duplicate data

- Data set may include data objects that are duplicates, or almost duplicates of one another
  - Major issue when merging data from heterogeous sources
- Example:
  - Same person with multiple email addresses
- Data cleaning
  - Includes process of dealing with duplicate data issues

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Why Data Preprocessing?

- No quality data, no quality mining results!

- Quality decisions must be based on quality data ‹
  e.g., duplicate or missing data may cause misleading statistics
- Data quality issues will influence the performance of some algorithms

- Data extraction, cleaning, and transformation comprises the majority of the work of building a data mining system

How?

- Aggregation and Sampling
- Discretization
- Feature transformation
- Dimensionality reduction
  - Choose subset of existing features
  - Create smaller number of new features

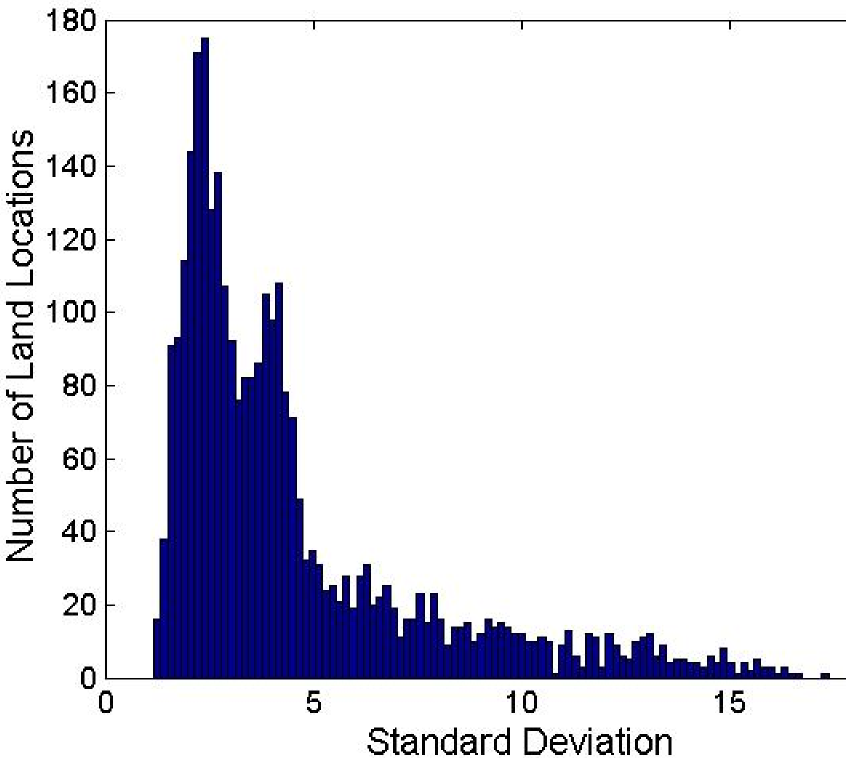
Aggregation

- Combining two or more attributes (or objects) into a single attribute (or object)

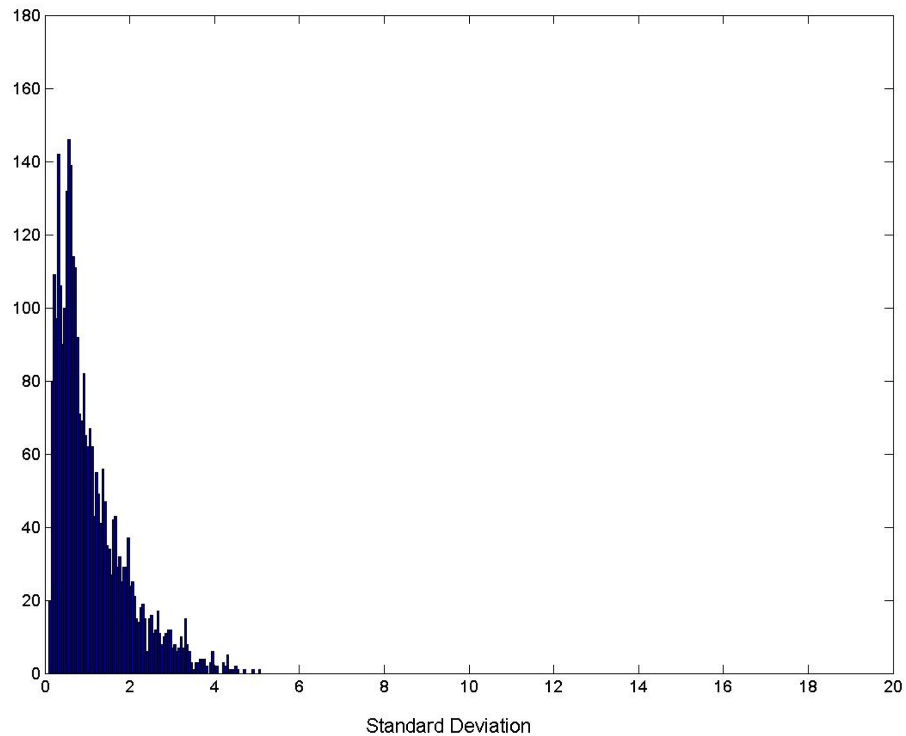
- Purpose
  - Data reduction
    - Reduce the number of attributes or objects
  - Change of scale
    - Cities aggregated into regions, states, countries, etc.
  - More “stable” data
    - Aggregated data tends to have less variability

Aggregation

Variation of precipitation in Australia



Standard deviation of average monthly precipitation



Standard deviation of average yearly precipitation

Sampling

- Sampling is the main technique employed for data selection.
  - Often used for both preliminary investigation of the data and the final data analysis.

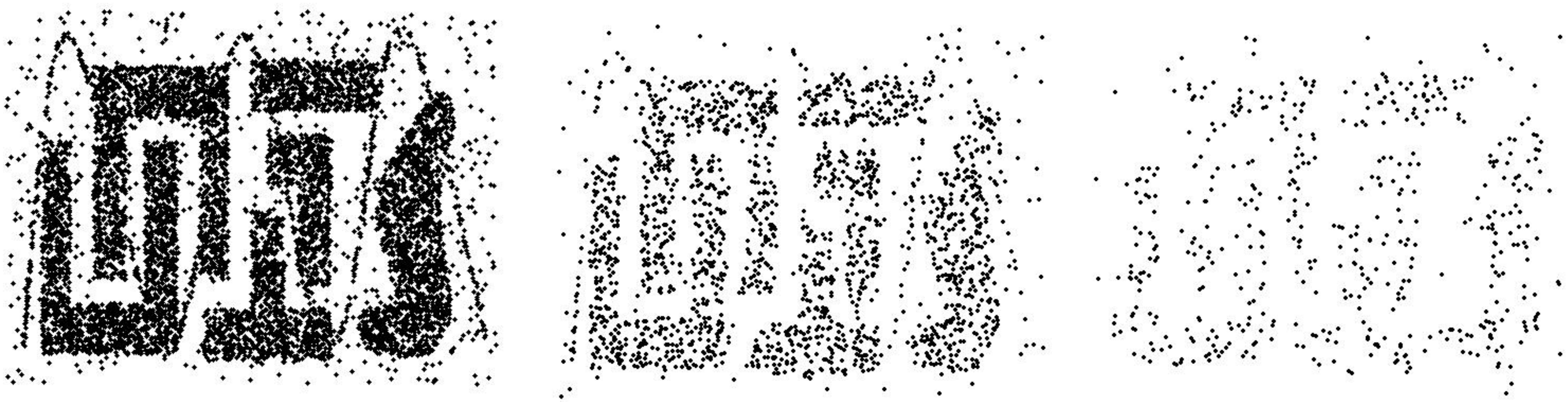
- Statisticians sample because obtaining the entire set of data of interest is too expensive or time consuming.

- Sampling is used in data mining because processing the entire set of data of interest is too expensive or time consuming.

Types of Sampling

- Simple random sampling
  - There is an equal probability of selecting any particular item.
- Sampling without replacement
  - As each item is selected, it is removed from the population.
- Sampling with replacement
  - Objects are not removed from the population as they are selected for the sample.
    - Same object can be selected more than once.
- Stratified sampling
  - Split the data into several partitions; then draw random samples from each partition.
  - Example: During polling, you might want equal numbers of male and female respondents. You create separate pools of men and women, and sample separately from each.

Sample size

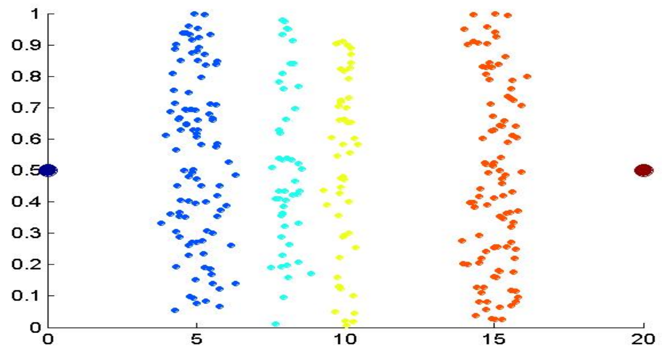


8000 points　　　　2000 Points　　　　500 Points

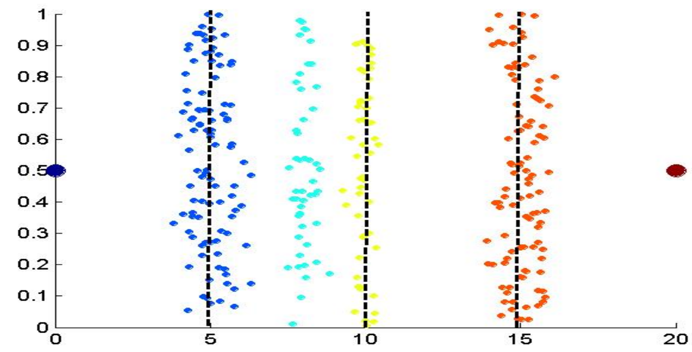
Discretization

- Discretization is the process of converting a continuous attribute into an ordinal attribute
  - A potentially infinite number of values are mapped into a small number of categories
  - Discretization is used in both unsupervised and supervised settings

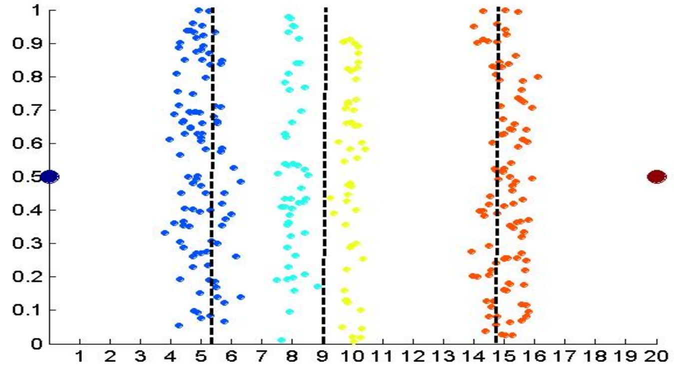
Approaches to Discretization



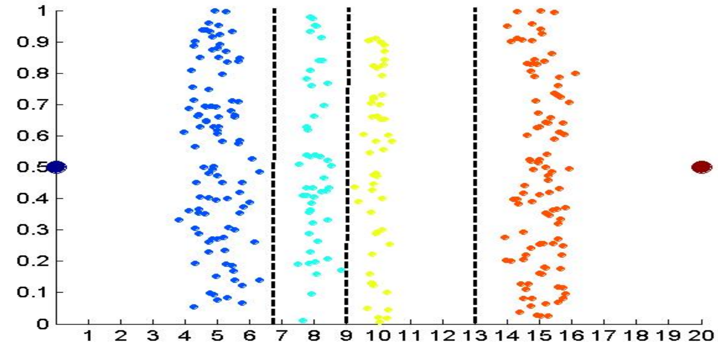
Original attribute values



Equal interval width



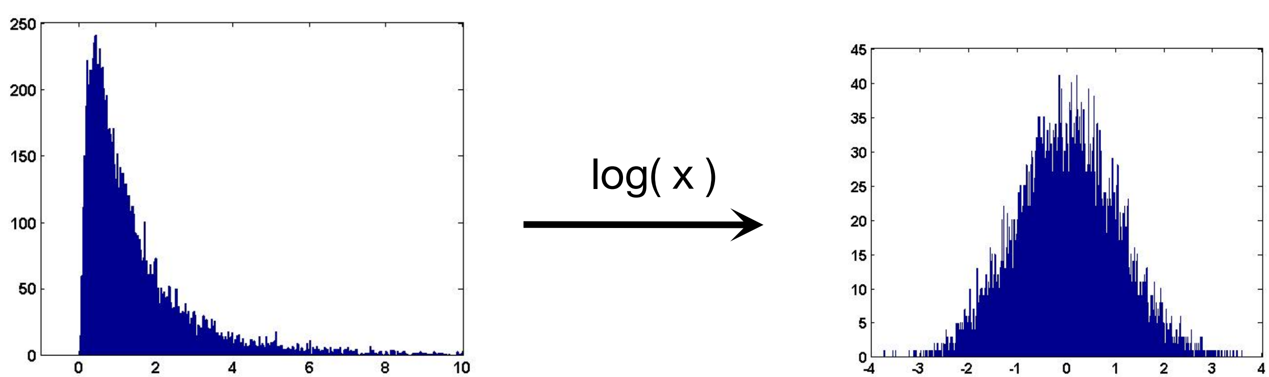
Equal frequency



k-means

Feature Transformation

- **Feature Transformation:** a function that maps the entire set of values of a given attribute to a new set of replacement values, such that each old value can be identified with one of the new values.
- Simple functions
  - Examples of transform functions:
    x^k log(x) e^x | x |
  - Often used to make the data more like some standard distribution, to better satisfy assumptions of a particular algorithm.



Feature Transformation

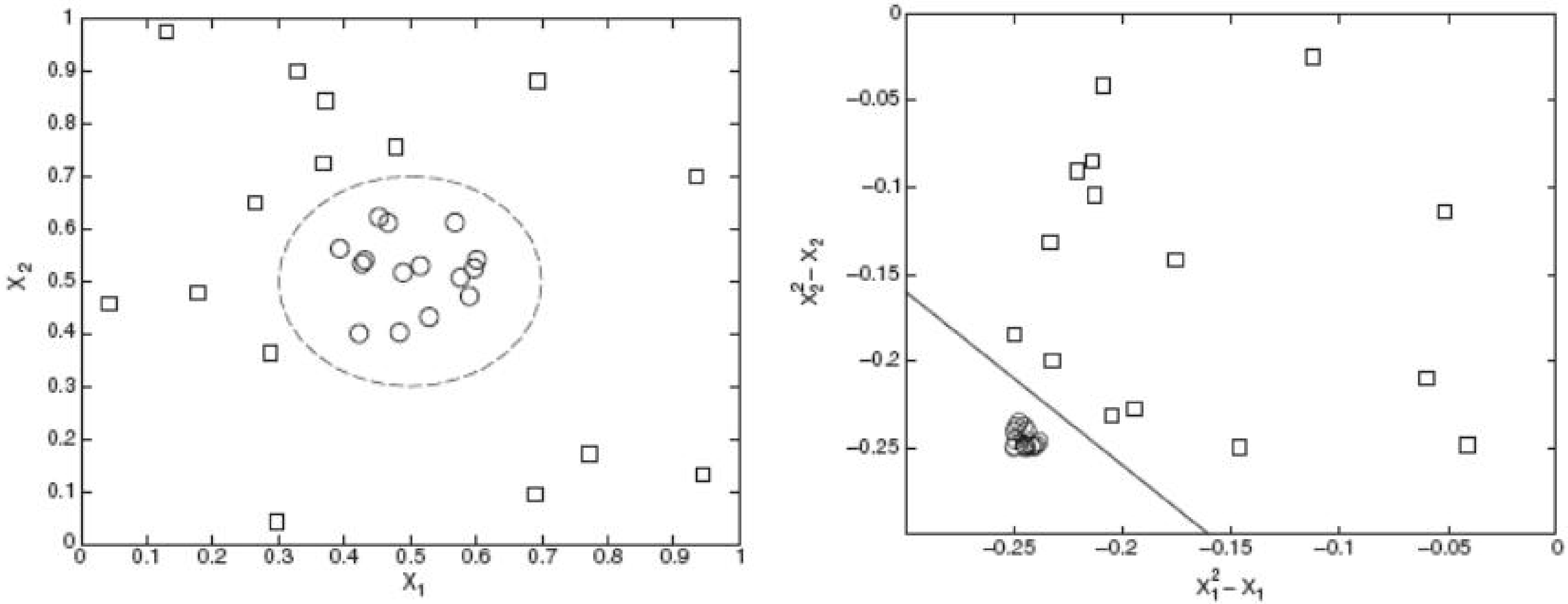
- Standardization or normalization
  - Usually involves making attribute:

mean	= 0
standard deviation	= 1

  - Important when working in Euclidean space and attributes have very different numeric scales.
  - Also necessary to satisfy assumptions of certain algorithms.
    - Example: principal component analysis (PCA) requires each attribute to be mean-centered (i.e. have mean subtracted from each value)

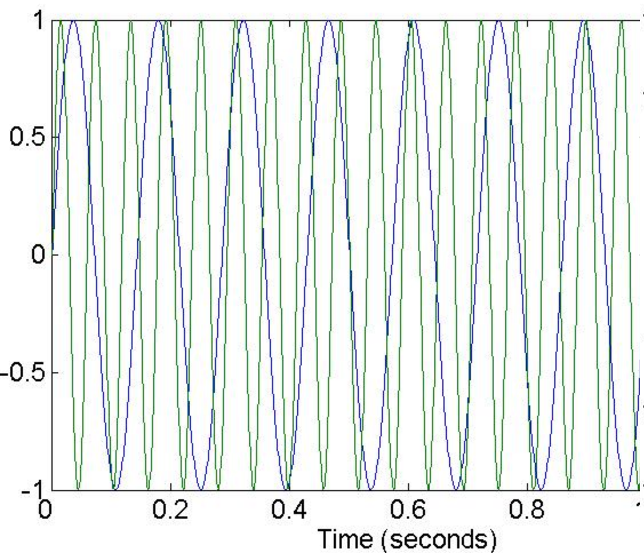
Transform Data to A New Space

$$\Phi:[x_1,x_2]\rightarrow[(x_1^{\,2}-x_1),(x_2^{\,2}-x_2)]$$

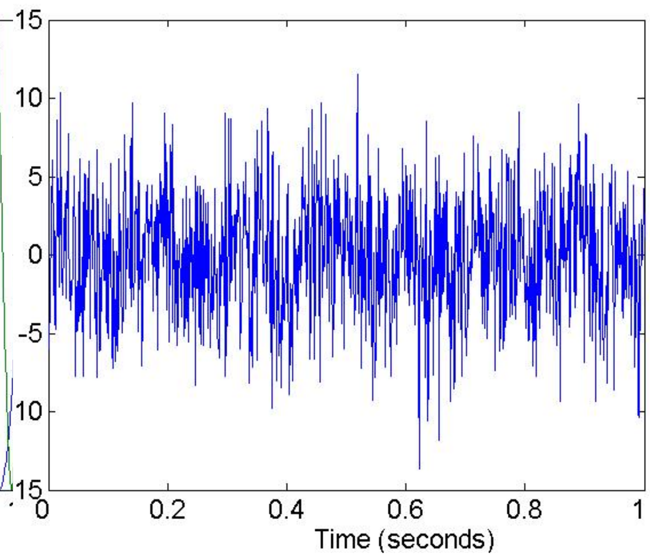


Transform Data to A New Space

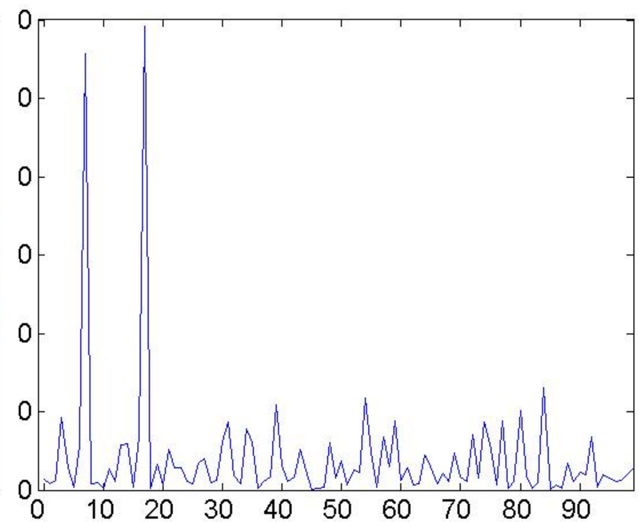
- Fourier transform
  - Eliminates noise present in time domain



Two sine waves



Two sine waves + noise



Frequency

Dimension reduction

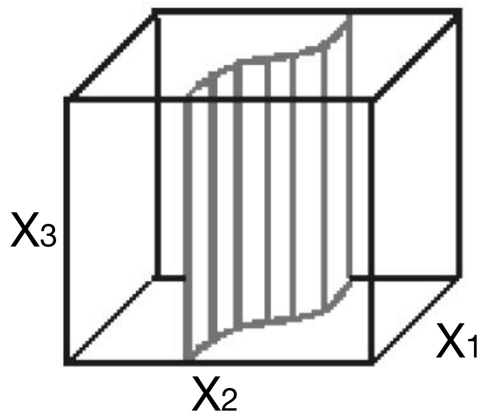
- Many modern data domains involve huge numbers of features / dimensions
  - Documents: thousands of words, millions of bigrams
  - Images: thousands to millions of pixels
  - Genomics: thousands of genes, millions of DNA polymorphisms

Why reduce dimensions?

- High dimensionality has many costs
  - Redundant and irrelevant features degrade performance of some ML algorithms
  - Difficulty in interpretation and visualization
  - Computation may become infeasible
  - Curse of dimensionality
- Two Methods
  - Feature selection
  - Latent feature creation

Feature Selection

- **Feature Selection** : Only a few features are relevant to the learning task



X3 - Irrelevant

Name	Math	English	Data Mining
001	90	100	98
002	90	58	90
003	90	32	91
004	90	68	92

Feature selection

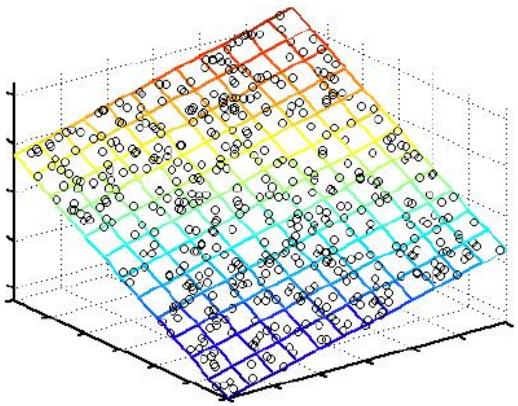
- Reduces dimensionality of data without creating new features
- Motivations:
  - Redundant features
    - highly correlated features contain duplicate information
    - example: purchase price and sales tax paid
  - Irrelevant features
    - contain no information useful for discriminating outcome
    - example: student ID number does not predict student's GPA
  - Noisy features
    - signal-to-noise ratio too low to be useful for discriminating outcome
    - example: high random measurement error on an instrument

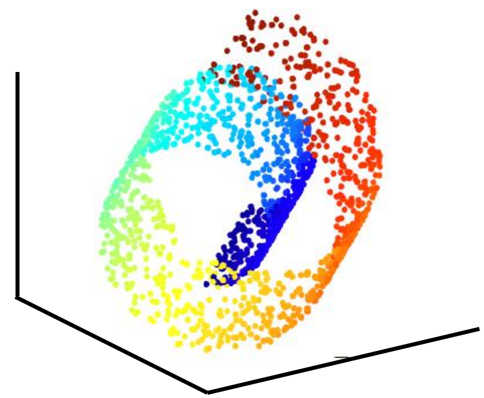
Feature selection: approaches

- Filter approaches:
  - Features selected before machine learning algorithm is run
- Wrapper approaches:
  - Use machine learning algorithm as black box to find best subset of features
- Embedded approaches:
  - Feature selection occurs naturally as part of the machine learning algorithm
  - Example: L1-regularized linear regression

Latent Features Creation

- **Latent features creation:** Some linear combination of features provides a more efficient representation than observed features



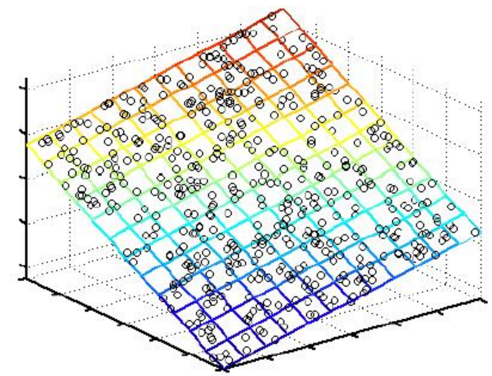


Name	Math	English	Data Mining
001	90	100	98
002	91	58	60
003	91	32	21
004	90	68	18

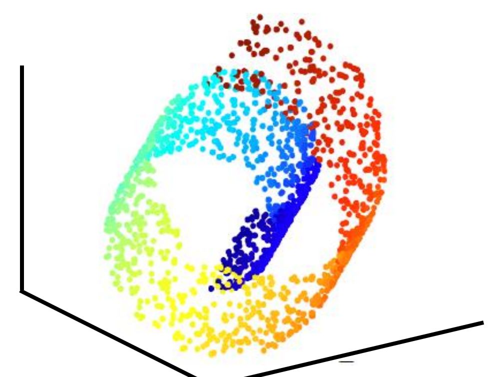
Latent Feature Creation

Motivation: Combinations of observed features provide more efficient representation, and capture underlying relations that govern the data

- Linear
  - **Principal Component Analysis (PCA)**
  - Factor Analysis
  - Independent Component Analysis (ICA)

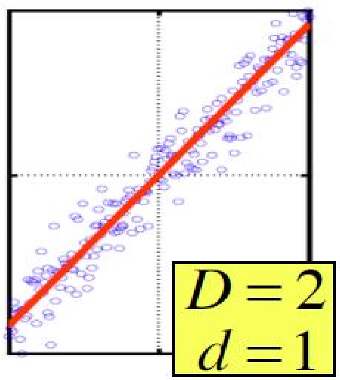


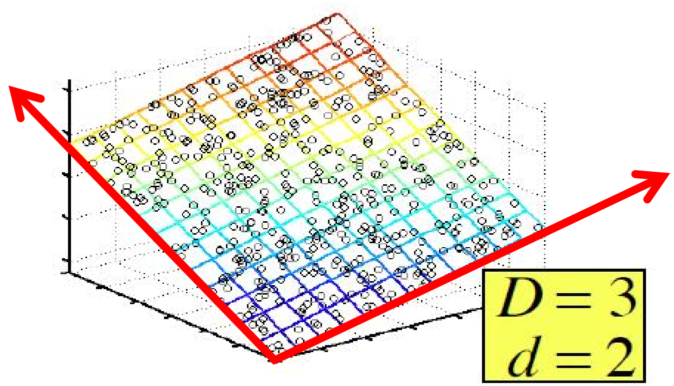
- Nonlinear
  - Laplacian Eigenmaps (拉普拉斯特征映射)
  - ISOMAP
  - Local Linear Embedding (LLE)



Principal Component Analysis

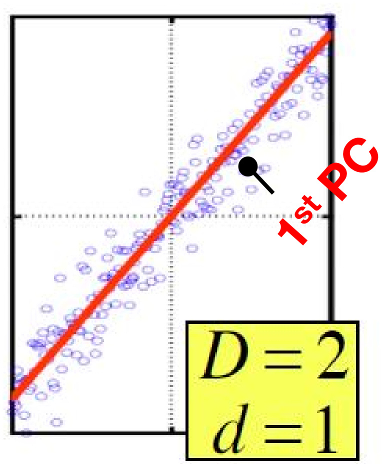
- **Assumption:** Data lies on or near a low *d*-dimensional linear subspace.
- Axes of this subspace are an effective representation of the data
- Identifying the axes is known as Principal Components Analysis, and can be obtained by Eigen or Singular value decomposition





Principal Component Analysis

- PCA produces a low-dimensional representation of a dataset.
- It finds a sequence of linear combinations of the features that have maximal variance, and are mutually uncorrelated.

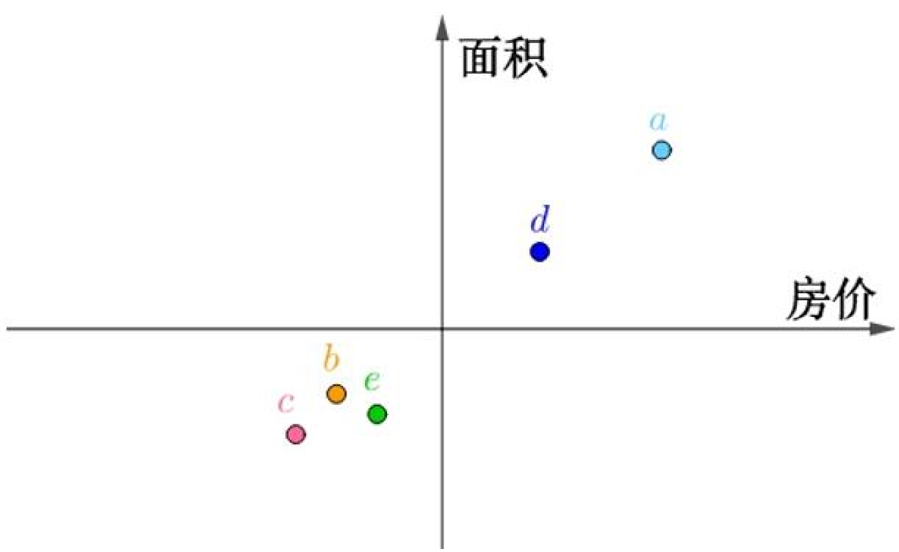


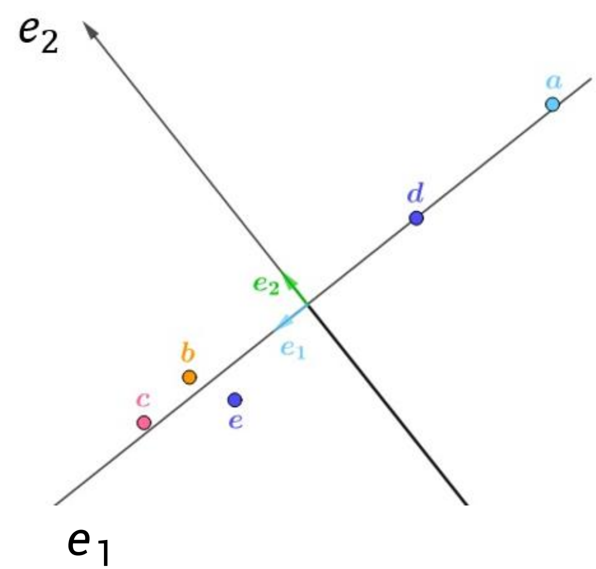
- Principal Components (PC) are orthogonal directions that capture most of the variance in the data
- 1st PC : direction of greatest variability in data
- 2nd PC : Next orthogonal (uncorrelated) direction of greatest variability
- And so on …

An Example

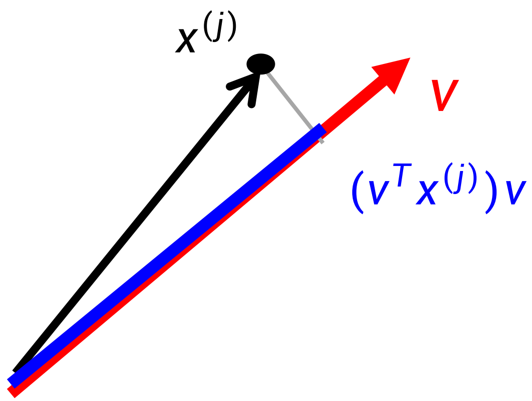
	房价(百万元)	面积(百平米)
a	5.4	4.4
b	−2.6	−1.6
c	−3.6	−2.6
d	2.4	1.9
e	−1.6	−2.1

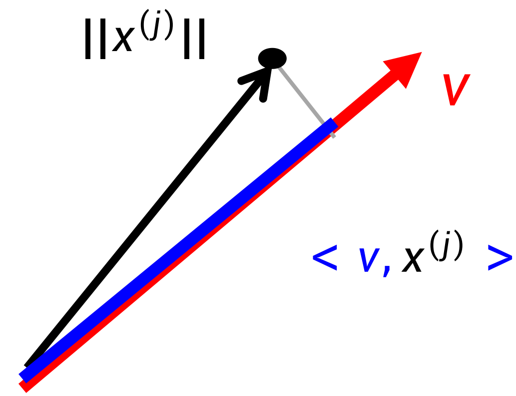
	主元1	主元2
a	−6.94	0.084
b	3.02	0.364
c	4.42	0.204
d	−3.05	−0.006
e	2.55	−0.646





Principal Component Analysis

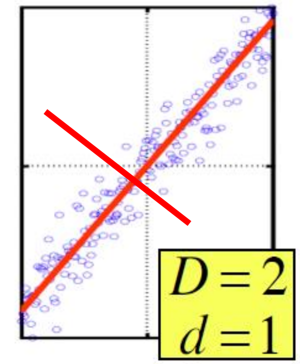




- Take a data point $x^{(j)}$ (D-dimensional vector)
- Projection of $x^{(j)}$ onto the 1st PC v is $(v^T x^{(j)})v$
- The distance on PC v is $(v^T x^{(j)})$
- v is always a unit vector
- If $x^{(j)}$ has been centered, then $\sum_{j=1}^{n} (\mathrm{v}^T \mathrm{x}^{(j)})^2$ represents the variance on PC v

Principal Component Analysis

- PCs: $v_1, v_2, \ldots, v_d,$



- Orthogonal and unit norm $\quad v_i^T v_j = 0 \quad i \neq j$

$$v_i^T v_i = 1$$

- Find vector that maximizes sample variance (scaled) of projection

Give the data, how to find PCs?

- Assume data are centered
- Data points $X = [x^{(1)}, \ldots x^{(n)}] \in R^{D \times n}$

Principal Component Analysis

$$\sum_{j=1}^{n}(\mathbf{v}^T\mathbf{x}^{(j)})^2 = \mathbf{v}^T\mathbf{X}\mathbf{X}^T\mathbf{v}$$

$$\max_{\mathbf{v}} \quad \mathbf{v}^T\mathbf{X}\mathbf{X}^T\mathbf{v} \quad \text{s.t.} \quad \mathbf{v}^T\mathbf{v} = 1$$

Wrap constraints into the objective function

$$L(v,\ \lambda) = V^T X X^T V - \lambda(V^T V - 1)$$

$$\begin{cases} \partial L(v,\lambda)/\partial\lambda = V^T V - 1 = 0 \\ \partial L(v,\lambda)/\partial v = 2\, X X^T\, V - 2\lambda V = 0 \end{cases}$$

$$\Rightarrow \boxed{(\mathbf{X}\mathbf{X}^T)\mathbf{v} = \lambda\mathbf{v}}$$

Principal Component Analysis

- **Therefore, v is the eigenvector of sample covariance matrix XX^T**

$$(\mathbf{X}\mathbf{X}^T)\mathbf{v} = \lambda\mathbf{v}$$

- **The eigenvalue λ denotes the amount of variability captured along that dimension**

Sample variance of projection = $\mathbf{v}^T\mathbf{X}\mathbf{X}^T\mathbf{v} = \lambda\mathbf{v}^T\mathbf{v} = \lambda$

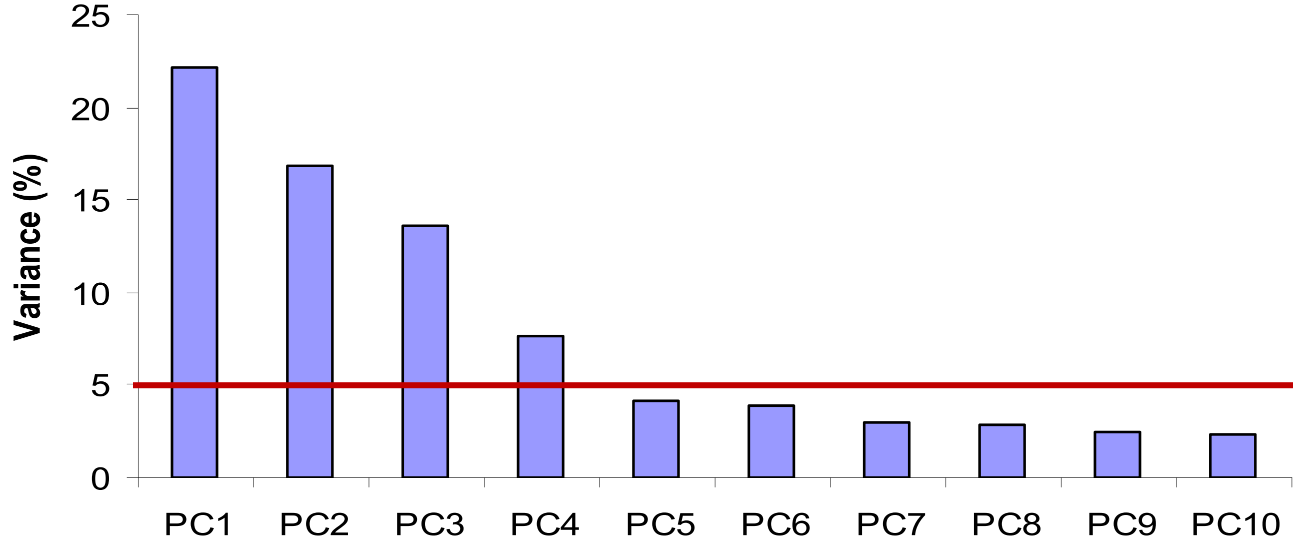
- **Eigenvalues** $\lambda_1 > \lambda_2 > \lambda_3 > \ldots$
  - The 1st Principal component v_1 is the eigenvector of XX^T associated with the largest eigenvalue λ_1
  - And so on …

Principal Component Analysis

- Original data: $x^{(j)} = \left[x_1^{(j)}, x_2^{(j)}, \ldots, x_D^{(j)}\right] \in R^D$
- Transformed data: $[v_1^T x^{(j)}, v_2^T x^{(j)}, \ldots, v_d^T x^{(j)}] \in R^d$ (reduced dim $d \ll D$)
- Transformed features are uncorrelated.

Dimensionality Reduction

In high-dimensional problem, data usually lies near a linear subspace. Only keep data projections onto principal components with **large** eigenvalues and *ignore* the components of lesser significance.



You might lose some information, but if the eigenvalues are small, you don't lose much

Total variance: $\sum_j Var(X^{(j)})$

Variance explained by the mth PC: $\frac{1}{n}\sum_j (v_m^T x^{(j)})^2$

Case Study: Bank Marketing

Bank Marketing Data

The data is related with direct marketing campaigns of a Portuguese banking institution. The marketing campaigns were based on phone calls. Often, more than one contact to the same client was required, in order to access if the product (bank term deposit) would be ('yes') or not ('no') subscribed. Variables are listed as follows:

Bank client data:

1 - age (numeric)
2 - job : type of job (categorical: 'admin.','blue-collar','entrepreneur','housemaid','management','retired','self-employed','services','student','technician','unemployed','unknown')
3 - marital : marital status (categorical: 'divorced','married','single','unknown'; note: 'divorced' means divorced or widowed)
4 - education (categorical: 'basic.4y','basic.6y','basic.9y','high.school','illiterate','professional.course','university.degree','unknown')
5 - default: has credit in default? (categorical: 'no','yes','unknown')
6 - housing: has housing loan? (categorical: 'no','yes','unknown')
7 - loan: has personal loan? (categorical: 'no','yes','unknown')

Related with the last contact of the current campaign:

8 - contact: contact communication type (categorical: 'cellular','telephone')
9 - month: last contact month of year (categorical: 'jan', 'feb', 'mar', ..., 'nov', 'dec')
10 - day_of_week: last contact day of the week (categorical: 'mon','tue','wed','thu','fri')
11 - duration: last contact duration, in seconds (numeric). Important note: this attribute highly affects the output target (e.g., if duration=0 then y='no'). Yet, the duration is not known before a call is performed. Also, after the end of the call y is obviously known. Thus, this input should only be included for benchmark purposes and should be discarded if the intention is to have a realistic predictive model.

Other attributes:

12 - campaign: number of contacts performed during this campaign and for this client (numeric, includes last contact)
13 - pdays: number of days that passed by after the client was last contacted from a previous campaign (numeric; 999 means client was not previously contacted)
14 - previous: number of contacts performed before this campaign and for this client (numeric)
15 - poutcome: outcome of the previous marketing campaign (categorical: 'failure','nonexistent','success')

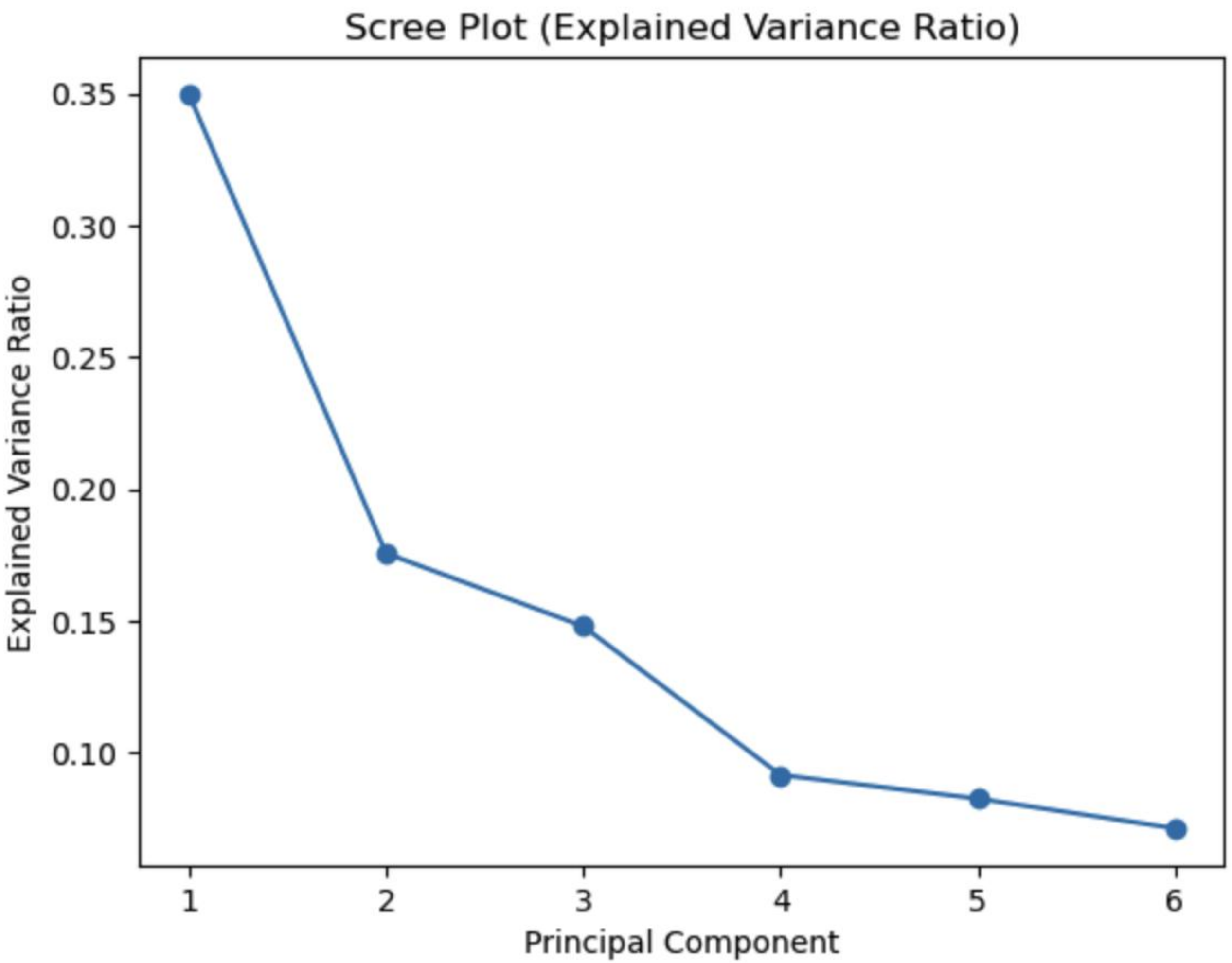
Social and economic context attributes

16 - emp.var.rate: employment variation rate - quarterly indicator (numeric)
17 - cons.price.idx: consumer price index - monthly indicator (numeric)
18 - cons.conf.idx: consumer confidence index - monthly indicator (numeric)
19 - euribor3m: euribor 3 month rate - daily indicator (numeric)
20 - nr.employed: number of employees - quarterly indicator (numeric)

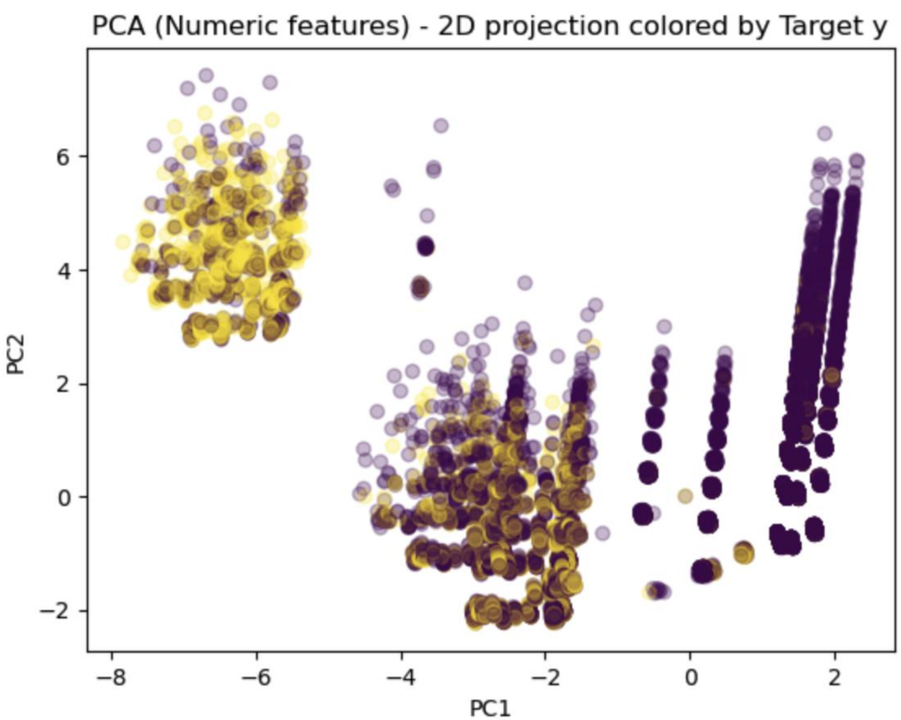
Desired target:

21 - y - has the client subscribed a term deposit? (binary: 'yes','no')

Case Study: Bank Marketing



Case Study: Bank Marketing



	feature	abs_loading	signed_loading
0	euribor3m	0.451812	0.451812
1	emp.var.rate	0.447702	0.447702
2	nr.employed	0.441803	0.441803
3	previous	0.336332	-0.336332
4	cons.price.idx	0.324057	0.324057
5	pdays	0.288998	0.288998
6	was_contacted_before	0.288986	-0.288986
7	campaign	0.095546	0.095546
8	cons.conf.idx	0.076774	0.076774
9	age	0.006284	-0.006284

	feature	abs_loading	signed_loading
0	contacts_total	0.553763	0.553763
1	campaign	0.492673	0.492673
2	was_contacted_before	0.377920	0.377920
3	pdays	0.377884	-0.377884
4	previous	0.224133	0.224133
5	cons.price.idx	0.222077	0.222077
6	emp.var.rate	0.163609	0.163609
7	euribor3m	0.136199	0.136199
8	cons.conf.idx	0.111612	0.111612
9	nr.employed	0.074192	0.074192

Some Takeaways

- Data quality issues
- Data preprocessing
- PCA

The End

THANK YOU

Any questions?